

Optimization Algorithms



CS4234

Week 4

LP Relaxation, Vertex Cover,
Maximum Satisfiability

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An ILP optimizes a linear objective with integer decisions

Variables

$x \in \mathbb{Z}^n$, often binary: $x_j \in \{0,1\}$.

Objective

maximize $c^T x$ (or minimize $c^T x$).

Constraints

$Ax \leq b$, where every expression is linear.

Integrity models discrete choices—but usually makes optimization hard.

An LP (linear program) relaxation allows fractional decisions

ILP

$$Ax \leq b$$

$$x_j \in \{0,1\}$$

Discrete feasible points

LP relaxation

$$Ax \leq b$$

$$0 \leq x_j \leq 1$$

Continuous feasible region

For maximization: $OPT_{LP} \geq OPT_{ILP}$ because the LP has every integer feasible solution plus more (fractional) feasible solutions.

A two-variable example shows the relaxation gap

maximize $3x_1 + 2x_2$

subject to

$2x_1 + x_2 \leq 2$

$x_1 + 2x_2 \leq 2$

ILP

$x_1, x_2 \in \{0, 1\}$

Best point: $(1, 0)$

value = 3

LP

$0 \leq x_1, x_2 \leq 1$

Best point: $(2/3, 2/3)$

value = $10/3$

Relaxation enlarges the feasible set, so its optimum can be strictly better.

LP relaxation follows a four-stage recipe



1

Model the problem as an ILP.

2

Relax integrality to obtain an LP.

3

Solve the LP in polynomial time.

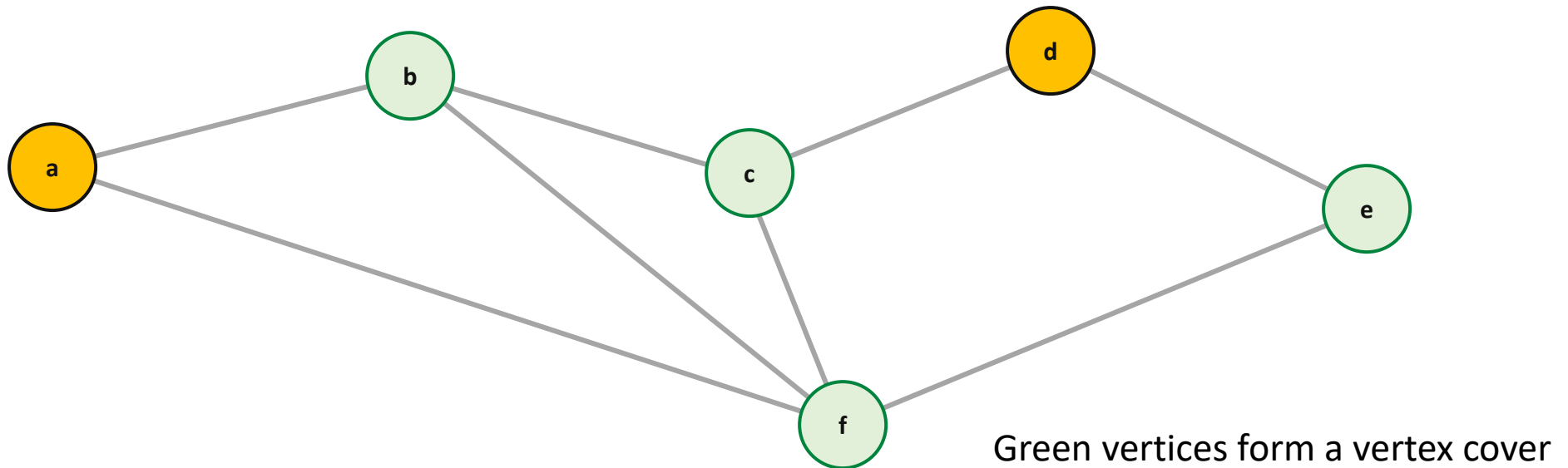
4

Round the fractional solution.

The analysis compares the rounded solution with the LP upper bound.

Vertex Cover: select vertices that touch every edge

A set $C \subseteq V$ is a vertex cover if every edge has at least one endpoint in C .



Goal: minimize $|C|$.

Vertex Cover ILP: variables select vertices

Variables

$x_v \in \{0,1\}$ for every vertex $v \in V$.

Objective

minimize $\sum_v x_v$

Constraints

$x_u + x_v \geq 1$ for every edge $\{u,v\} \in E$.

$x_v = 1$ selects v ; each edge requires at least one selected endpoint.

Threshold rounding gives a 2-approximation

- 1 Solve the LP and obtain x^* .
- 2 Select every vertex v with $x_v^* \geq 1/2$.
- 3 Return the selected set C .

Feasible

$$x_u^* + x_v^* \geq 1 \Rightarrow \text{endpoint} \geq 1/2$$

Cost

$$\begin{aligned} |C| &\leq 2 \sum_v x_v^* \\ &= 2 \cdot \text{OPT}_{\text{lp}} \leq 2 \cdot \text{OPT} \end{aligned}$$

Threshold 1/2 covers every edge and increases the LP cost by at most factor 2.

Max-SAT asks us to satisfy as many clauses of a CNF as possible

Literal

A Boolean variable x_j or its negation $\neg x_j$.

Clause

An OR of literals, e.g. $C_i = x_1 \vee \neg x_2 \vee x_4$.

CNF

An AND of clauses: $\Phi = C_1 \wedge C_2 \wedge \cdots \wedge C_m$.

Max-SAT

Choose an assignment maximizing the satisfied clause weight.

SAT asks whether all clauses can be satisfied; Max-SAT optimizes how many.

Weighted Max-SAT uses clause values

Weight

Each clause C_i has nonnegative weight w_i .

Indicator

$\tau(C_i) = 1$ if the assignment satisfies C_i ; otherwise 0.

Objective

$\text{cost}(\tau) = \sum_i w_i \tau(C_i)$.

Goal

Choose τ maximizing $\text{cost}(\tau)$.

Linearity of expectation also works for weighted clauses.

Max-SAT ILP: variables record truth and satisfaction

Truth

$x_j \in \{0,1\}$: whether Boolean variable x_j is true.

Clause

$y_i \in \{0,1\}$: whether clause C_i is satisfied.

Objective

maximize $\sum_i w_i y_i$.

The objective pays for each clause whose satisfaction variable reaches 1.

One linear constraint certifies each satisfied clause

Consider a clause C_i .

$$\sum_{j: x_j \in C_i} x_j + \sum_{j: \bar{x}_j \in C_i} (1 - x_j) \geq y_i$$

Together with $x_j, y_i \in \{0,1\}$, y_i can be 1 only if at least one literal is true.

The ILP is exact: every Boolean assignment corresponds to a feasible integer solution.

Relax, solve, and round the Max-SAT ILP (let OPT is optimal ILP)

1

Replace $x_j, y_i \in \{0,1\}$ by $x_j, y_i \in [0,1]$.

2

Solve the LP to get x^* and y^* .

3

Set $x_j = \text{true}$ independently with probability x_j^* .

4

Compare the rounded value with $OPT_{LP} \geq OPT$.

Fractional variables become probabilities for randomized rounding.

Randomized rounding uses the LP values as coin biases

- 1 Solve the LP and obtain x_j^* for every variable.
- 2 Independently set $x_j = \text{true}$ with probability x_j^* .
- 3 Evaluate and return the resulting assignment τ .

LP value:
 $OPT_{LP} = \sum_i w_i y_i^*$

The LP solution guides a nonuniform random assignment.

A clause fails only if every literal rounds false

For clause C_i :

$\Pr[C_i \text{ not satisfied}] = ?$

Independence turns the joint failure event into a product.

Clause analysis again reduces to one local probability.

A clause fails only if every literal rounds false

For clause C_i :

$$\Pr[C_i \text{ not satisfied}] = \prod_{j: x_j \in C_i} (1 - x_j^*) \prod_{j: \bar{x}_j \in C_i} x_j^*$$

Independence turns the joint failure event into a product.

Clause analysis again reduces to one local probability.

Expected clause value is one minus the failure product

$$\mathbb{E}[\tau(C_i)] = 1 - \prod_{j: x_j \in C_i} (1 - x_j^*) \prod_{j: \bar{x}_j \in C_i} x_j^*$$

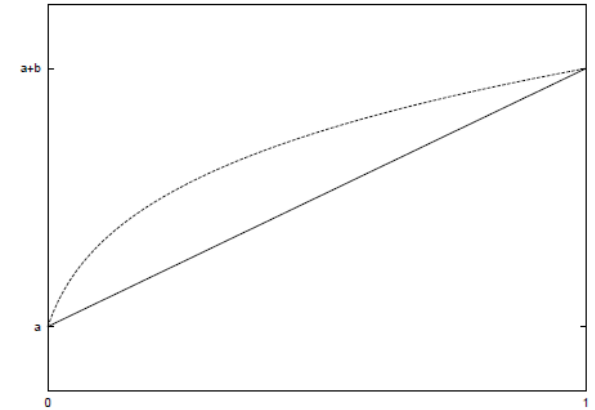
We now upper-bound the failure product using the LP constraint.

The randomized-rounding guarantee is hidden inside this product.

Analyzing Randomized Rounding

- **AM-GM inequality:** For non-negative $a_1, a_2, \dots, a_k$

$$\left(\prod_i a_i \right)^{1/k} \leq \frac{1}{k} \sum_i a_i$$



- **Fact on Concave Function:** For any concave function $f(x)$ on $[0,1]$, suppose $f(0) = a$ and $f(1) = a + b$, then $f(x) \geq bx + a$ for $x \in [0,1]$

Analyzing Randomized Rounding

Analyzing Randomized Rounding

- $\mathbb{E}[\tau(C_i)] = 1 - \prod_{j:x_j \in C_i} (1 - x_j^*) \prod_{j:\overline{x_j} \in C_i} x_j^*$ (C_i has ℓ_i literals)
 $\geq 1 - \left(\frac{1}{\ell_i} (\sum_{j:x_j \in C_i} (1 - x_j^*) + \sum_{j:\overline{x_j} \in C_i} x_j^*) \right)^{\ell_i}$ (By AM-GM)
 $= 1 - \left(1 - \frac{1}{\ell_i} \left(\sum_{j:x_j \in C_i} x_j^* + \sum_{j:\overline{x_j} \in C_i} (1 - x_j^*) \right) \right)^{\ell_i}$
 $\geq 1 - \left(1 - \frac{y_i^*}{\ell_i} \right)^{\ell_i}$ (By LP constraint)
 $\geq \left(1 - \left(1 - \frac{1}{\ell_i} \right)^{\ell_i} \right) y_i^*$ (By concave function fact)
 $\geq \left(1 - \frac{1}{e} \right) y_i^*$

The LP constraint controls the clause failure probability

For clause C_i containing ℓ_i literals

$$E[\tau(C_i)] = 1 - \left(1 - \frac{y_i^*}{\ell_i}\right)^{\ell_i}$$

$$\text{Hence } E[\tau(C_i)] \geq \left(1 - \left(1 - \frac{1}{\ell_i}\right)^{\ell_i}\right) y_i^* \leq \left(1 - \frac{1}{e}\right) y_i^*$$

Concavity turns the nonlinear clause probability into a linear fraction of y_i^* .

Summing clauses gives a $1-1/e$ guarantee

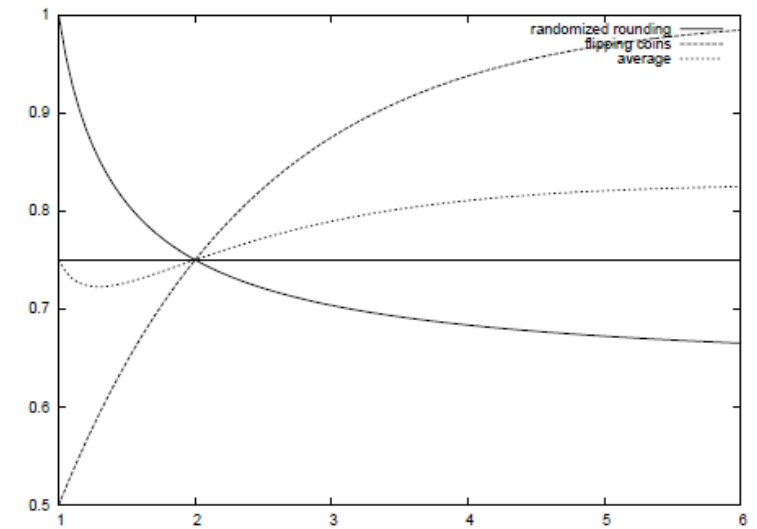
$$\mathbb{E}[\text{cost}(\tau)] \geq (1-1/e) \sum_i w_i y_i^* = (1-1/e) \text{OPT}_{LP} \geq (1-1/e) \text{OPT}$$

Numerically, $1-1/e \approx 0.632$.

LP relaxation plus randomized rounding is a 1.582-approximation for Max-SAT.

Combining two algorithms reaches factor 3/4

Output Best of the following two:
Algo1 (flipping unbiased coins), and
LP with randomized rounding



Summing over clause weights gives $E[\text{cost}] \geq \frac{3}{4} \cdot \text{OPT}$.

Three techniques recur throughout approximation algorithms

Randomize

Use indicators and expectation to analyze simple local choices.

Derandomize

Preserve conditional expectation while fixing decisions.

Relax + round

Solve a tractable fractional problem, then convert it to a discrete solution.

Design the algorithm around a bound that can pay for every rounding loss.

References

- **Max-SAT:**

- Chapter 5 of [Williamson and Shmoys]

- Chapter 16 of [Vazirani]